

Measuring and Recording the Tide in the Shark River

Using Amateur Electronics to Explore our Local Environment

Tides play an important role in the marine life along our shores; understanding what influences tides and then using this knowledge to predict tide levels is important not only to monitoring marine ecology, but for navigation and boating safety as well.^{1,2} Our Shark River Estuary is a unique waterway a stone's throw away from Info Age. Measuring the tides offers us the opportunity to learn more about it and to connect with students interested in science, technology, and our environment.

The National Oceanic and Atmospheric Administration (NOAA) maintains tide sensors along US coast, including one in the Shark River Inlet. These instruments use sensors that measure the changing water surface level from a fixed point on a pier or bridge. They are built to withstand harsh environments, but are relatively large, and require external power and a satellite network to function.³

At the InfoAge Maker Space, the CDL, we are developing a program to build and deploy tide sensors, connected by radio, to record the changing tide level at several locations of the Shark River estuary throughout the year. We plan to collect this data at a receiver at Camp Evans in order to explore how the moon, sun, wind speed and direction, and rainfall influence the changing tides.

The sensors we are designing are to be robust, independently operating, and unobtrusive:

- Robust - these will be inexpensive, low maintenance units that use ultrasonic or lidar sensors to measure water levels and that require essentially no moving parts. We will create weather resistant 3D printed containers to house them.
- Independently Operating - we will use small solar panels and batteries to power them. Power management of the microcontroller, sensor and transmitter electronics is critical so that we don't need large arrays and storage to operate 24 hours a day. The LoRa radio and the Meshtastic network offer a way to instantly transmit sensor data around the river instead of recording the data locally.

- Unobtrusive - we will require the permission of others to place these units along docks, so “staying out of the way” is extremely important. Besides making sure we don’t have to constantly repair them or require power hookups, the sensors must be physically small and be housed in containers that do not create concern. Other published reports do not offer examples that reach all of these goals.^{4,5,6,7}

The CDL members bring expertise in electronics, radio, 3D printing, software development, and design to bear on this project. It offers an opportunity to create education programs where students can explore a wide variety of scientific and engineering topics including radio communications, electronics, marine ecology, data analysis, and machine learning.

1) Garrett, Chris, and Leo Maas. "Tides and their effects." *Oceanus* 36, no. 1 (1993): 27-37.

2) <https://boaterusa.com/understanding-tides-and-currents/>

3) https://oceanservice.noaa.gov/education/tutorial_tides/tides11_newmeasure.html

4) Haq, Nabil Amirul, Khomsin, and Danar Guruh Pratomo. "The design of an Arduino based low-cost ultrasonic tide gauge with the internet of things (iot) system." In *IOP Conference Series: Earth and Environmental Science*, vol. 698, no. 1, p. 012004. IOP Publishing, 2021.

5) Kusuma, Andi, Viv Djanat Prasita, and Robert Tambun. "IoT-Based Real-Time Tide Monitoring Tool: Design and Case Study at Kenjeran Beach, Surabaya." *Transactions on Maritime Science* 14, no. 1 (2025).

6) J. L. Rocha et al., "Wave Profile and Tide Monitoring System for Scalable Implementation," 2023 IEEE SENSORS, Vienna, Austria, 2023, pp. 1-4, doi: 10.1109/SENSORS56945.2023.10325051.

7) Ginanjar, S., Adiningsih, S., Ikhtiarino, S. et al. An internet of things architecture for real-time tide and wave monitoring using Heltec LoRa 32 and Kalman filter. *J. Ocean Eng. Mar. Energy* 9, 219–231 (2023). <https://doi.org/10.1007/s40722-022-00258-7>.